

VENKAT MARELLA

Tech Lead & DevSecOps Architect | MLOps & LLMOps | Multi-Cloud Kubernetes

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PROFESSIONAL SUMMARY

Accomplished Tech Lead & DevSecOps Architect with 16 years of IT experience and 9+ years of enterprise experience in cloud infrastructure, site reliability engineering, and high-availability application delivery, currently advancing expertise across the complete MLOps, LLMOps, AIOps, and AI Agentic Operations lifecycle. Proven background in architecting, deploying, and managing production-grade Kubernetes platforms across Google Kubernetes Engine (GKE), Amazon EKS, Azure AKS, and Red Hat OpenShift.

Focuses on bridging enterprise platform engineering with cutting-edge AI systems, leveraging strong foundations in Linux automation, Python, Docker, Infrastructure as Code (Terraform, Ansible), and CI/CD pipelines to build scalable, self-healing, and resilient cloud-native AI platforms.

TECHNICAL STACK & CORE COMPETENCIES

- **Multi-Cloud Kubernetes & Infra:** Google Kubernetes Engine (GKE), Amazon EKS, Azure AKS, Red Hat OpenShift, Helm, Terraform, Ansible Playbooks, AWS (EC2, CloudFormation, IAM), GCP, Azure, Multi-Cloud Architecture.
- **DevOps & CI/CD Foundations:** Linux Shell Scripting, Python Automation, Git Version Control, Docker & Containerization, Jenkins, GitHub Actions, Microservices Security.
- **MLOps Pipelines & Engineering:** Data & Feature Engineering (Feast), Experiment Tracking (MLflow, Weights & Biases), Model Versioning (DVC), Model Deployment (ONNX, FastAPI, Batch Inference), Continuous Training, Model Monitoring & Drift Detection, ML Orchestration (Kubeflow, SageMaker, Vertex AI pipelines).
- **LLMOps & RAG Systems:** Prompt Engineering & Versioning, LLM Fine-Tuning (LoRA, QLoRA), Vector Databases (ChromaDB, Pinecone, Weaviate), RAG Architecture, Model Context Protocol (MCP), Quantization, LLM Observability & Guardrails.
- **AIOps & Self-Healing Infrastructure:** Telemetry & APM Integration, Statistical & ML-Based Anomaly Detection, Predictive Analytics & Capacity Planning, Root Cause Analysis, Auto-Remediation & Chaos Engineering, Kubernetes Microservices Observability.
- **AI Agentic Ops & Autonomous Systems:** Multi-Agent Systems, Agent Frameworks (LangChain, LlamaIndex, CrewAI), MCP Server/Client Integrations, Tool Use & Function Calling, Agent Memory (Vector DBs/Knowledge Graphs), Chain-of-Thought Reasoning, Human-In-The-Loop Patterns.
- **Monitoring, Observability & Systems:** Prometheus, Grafana, ELK/EFK Stack, Distributed Tracing, J2EE Application Server Migration (WebLogic, JBoss, WebSphere), DevSecOps Automation.

PROFESSIONAL EXPERIENCE

BenchPrep

Feb 2022 – Present

Chicago, IL

Tech Lead & DevSecOps / SRE Architect

- Lead and mentor the platform and DevSecOps engineering team while remaining a hands-on architect for production operations, Kubernetes platforms, cloud systems, and security initiatives.
- Spearhead AI Agentic Ops and AI-assisted DevOps practices by integrating Claude, OpenAI Codex, and Model Context Protocol (MCP) servers into daily SRE, Terraform, Python automation, and Kubernetes operations.
- Design MCP-powered workflows connecting Kubernetes, Terraform, New Relic, GitHub, and cloud platforms to automate troubleshooting, infrastructure analysis, and operational tasks.
- Build Python, Terraform, and Bash automation for Kubernetes operations, incident response, platform migrations, infrastructure provisioning, and production troubleshooting.
- Accelerate operational investigations by using LLMs to analyze Kubernetes events, application logs, infrastructure state, Terraform configurations, and observability data, reducing MTTR and engineering overhead.
- Design and architect highly available cloud infrastructure across AWS and IBM Cloud with automated scaling, ingress management, and disaster recovery.
- Implement and operate GitOps-based continuous delivery using Argo CD, improving deployment consistency and operational reliability across production microservices.
- Lead Kubernetes networking, ingress, security hardening, zero-downtime cluster upgrades, virtual machine, bare-metal, vendor, and cross-functional platform initiatives.

BenchPrep

Oct 2019 – Feb 2022

Chicago, IL

Senior DevOps / SRE Engineer

- Architected and administered container technology and Kubernetes cluster setups for microservices in IBM Cloud (IKS).
- Automated infrastructure provisioning with Terraform and Ansible while supporting AWS EC2, S3, IAM, and DynamoDB components.
- Migrated Helm 2.x templates to Helm 3.x without downtime and established Helm-driven automated release workflows.
- Created CI/CD pipelines using GitHub, Codeship, Ansible, and Terraform to support agile engineering delivery.
- Configured New Relic Kubernetes alerts to Slack, shipped centralized logs to Sematext cloud, and deployed Threat Stack / Lace Work FortiCNAPP security agents.
- Operated Kubernetes node autoscalers and Horizontal Pod Autoscaling (HPA) to maintain high availability under dynamic production traffic.

Novisys LLC (Client: United Airlines)

Jul 2017 – Oct 2019

Senior DevOps / Middleware Engineer

Chicago, IL

- Designed and implemented Docker Enterprise (DDC/UCP/DTR), Docker Swarm, and Red Hat OpenShift container platforms across AWS and on-prem data centers.
- Automated AWS infrastructure using Terraform and Ansible across EC2, EFS, S3, IAM, Lambda, RDS, security groups, and bastion architectures.
- Built CI/CD pipelines integrating Jenkins, TFS, TeamCity, CA Release Automation, AWS CodeDeploy, Docker Compose, and Kubernetes Helm templates.
- Configured Datadog monitoring and alert policies for production OpenShift clusters, and maintained OpenShift RPM and Red Hat Docker repositories in Artifactory.
- Conducted enterprise POCs for container adoption across ECS, EKS, OpenShift, Kubernetes, and Docker Enterprise.

Novisys LLC (Client: United Airlines)

Jan 2013 – Jun 2017

Senior Middleware Engineer

Chicago, IL

- Managed end-to-end lifecycle of WebLogic, WebSphere, and JBoss application server environments from development through mission-critical production.
- Automated deployments, domain provisioning, and component configurations using WLST, wsadmin, UNIX shell scripts, ANT, Jython, and Python.
- Configured and tuned JDBC data sources, JMS messaging, custom authenticators, F5 Big-IP load balancing, clusters, and Oracle 11g RAC integrations.
- Performed advanced production support, thread dump analysis, memory leak detection, SiteScope monitoring, and high-availability operations.

Novisys LLC (Client: United Airlines)

Jun 2010 – Dec 2012

Ecommerce Middleware Engineer

Chicago, IL

- Supported high-traffic ecommerce production infrastructure, WebLogic/WebSphere/JBoss application servers, and Apache/IIS web servers.
- Administered Oracle RAC JDBC connections, JMS queues, F5 load balancing, and third-party integrations (Unica, Endeca).
- Automated scheduled backups and monitoring scripts; conducted JVM performance tuning and root cause analysis during production incidents.